MatchPulse — Architecture Characteristics

This is the Task 2 source. It selects the few qualities that matter, links each to the broad requirements R1–R8, and states the level the system needs. Evidence and measurement detail stay in the linked records.

1. Selection

I use the course's 9+1+1 list and add **reliability** and **security** because the course treats them as operational and implicit characteristics. I keep five. The test: if this quality fails, does a documented business rule break, or can I no longer trust the system? If not, it is a design consideration, not a primary target.

Characteristic	R points	Why it matters	Needed level (1–5)	Quantified target
Reliability	R2, R3, R5, R7	Capacity, balances, positions, tickets and settlement are concurrency rules. A failure is silent and corrupts the ledger.	**5** — release-critical	100 concurrent requests against capacity 32: exactly 32 accepted, 68 refused for a business reason, 0 transport errors, committed state matches the counter. No negative balance. Every winning share paid once. A retry with the same key and payload is replayed, not charged twice.
Security	R1, R2, R3, R4, R7, R8	Staff-only management, the points ledger, attendee identity and club isolation create a trust boundary. The UI cannot be the boundary.	**5** — release-critical	0 anonymous private reads or mutations; 0 cross-club reads; 0 direct table writes; only `OWNER` / `ADMIN` / `STAFF` resolve a club market; `ANALYST` sees aggregates only; no service-role key in the app runtime.
Testability	R2, R3, R5, R7	The core rules are concurrency and authorization rules; inspection cannot prove them.	**5** — release-critical	Every critical invariant has a database-backed test that has been observed failing under a deliberate mutation; the full suite runs on every pull request in under 3 minutes.
Deployability	R1–R8	A one-person project across two course deadlines cannot spend days on infrastructure.	**3** — measured, not a production gate	Clean checkout to a running, verified local/CI environment in under 15 minutes; zero manual database steps beyond environment values; migrations replay from an empty schema.
Elasticity	R2, R4, R5	Match traffic is a step change at kickoff, not a gradual ramp.	**3** — measured, not a production gate	Under 100 concurrent registrations, p95 < 1,000 ms in the named CI environment, with no unhandled transport errors; after the burst, no unbounded work or connections.

Scale

1 = not needed; 2 = best effort; 3 = target defined and measured manually; 4 = measured in CI, not a release gate; 5 = release-critical and backed by observed-red evidence.

2. What each target means

Reliability. The system must be correct when many people act at once. A capacity refusal is a business answer, not an error. The same rule applies to ticket sales, points, positions, ticket codes and resolution. The target is stated per burst because "correct under load" without a load is not measurable.

Security. Authorization is a database property, not a hidden button. The target names the zeroes that matter: no anonymous private access, no cross-club read, no direct write, no service-role key in the runtime. Role and entitlement checks are part of the target because a club analyst must not see attendee identity and a canceled plan must not grant features.

Testability. A passing test is not evidence unless it can fail. The target is a failable, database-backed test for each critical rule, run on every change. The suite must stay under three minutes so the feedback loop remains useful.

Deployability. The target is reproducibility for one developer and one CI runner: a clean checkout, one local stack, one migration replay, one build, one health check. It is not a production availability target.

Elasticity. The target is the match-time burst, not steady growth. p95 is measured in the named CI environment; the post-peak resource release is part of the target but is not claimed without a deployed environment.

3. Deliberately not selected

- **Performance** separately: the burst p95 is covered by elasticity.
- **Scalability**: the kata states match-time peaks, not growth beyond one deployment.
- **Fault tolerance**: no hardware-failure or multi-node availability target.
- **Availability**: no hosted service, so an availability target would be fiction.
- **Modularity, simplicity, maintainability, evolvability**: design choices and consequences of the modular monolith, not separate success drivers.
- **Overall cost and partitioning type**: constraints and design dimensions, not qualities the system exhibits.

4. Agile note

These five characteristics are the **smallest useful set**, not a scorecard. They act as fitness functions: the database tests for reliability and security, the fault-injection matrix for testability, the CI delivery job for deployability and the burst measurement for elasticity. If the evidence changes, the target changes through a decision record, not by silently editing this table.

5. Detail

- Enforcement and test status: invariant registry
- Fault-injection matrix: verification record
- Delivery and recovery: verification record
- Broad and detailed requirements: requirements.md